

DSC 335, Handout 03: Bayes' Rule for Random Variables (*Chapter 2.3-2.4*)

LEARNING OBJECTIVES.

- *Calculate posterior probabilities using Bayes' Rule for random variables.*

Recall: Bayes' Rule for Events

For events A and B, the posterior probability of B given A is given by

$$P(B|A) = \frac{P(B)P(A|B)}{P(A)} = \frac{P(B)L(B|A)}{P(A)}.$$

Definition: A random variable is

Example 1. Give examples of discrete and continuous random variables.

Definition: The probability mass function (pmf) of a discrete random variable is

The Binomial distribution (model)

Let the random variable Y be the number of successes in a fixed number of trials n . Assume that the trials are independent and that the probability of success in each trial is π . Then the conditional dependence of Y on π can be modeled by a Binomial distribution (model) with parameters n and π :

In R: For a binomial random variable X , the function `dbinom(x, size, prob)` gives $P(X = x)$ and the function `pbinom(x, size, prob)` gives $P(X \leq x)$.

Example 2. Suppose you roll five six-sided dice (as in the game Yahtzee). What is the probability that you get the number 2 on four of the dice? more than two of the dice?

Recall: To build a Bayesian model, you need a prior, a likelihood, and a normalizing constant.

Example 3. Suppose you are playing a computer game that has three levels of difficulty: easy, medium, and hard. You have a 30% chance of beating the game on hard, a 60% chance of beating the game on medium, and a 90% chance of beating the game on easy. However, you do not get to choose the difficulty setting. Each day, the computer randomly chooses the difficulty setting for all games played that day. You plan on playing the game 5 times in a day, and based on your results you want to try to figure out what setting it was on.

- (a) Let π be the probability of beating the game. Since π changes based on the difficulty settings, it is a random variable. Suppose the computer is equally likely to choose any difficulty setting. Give the pmf for π . (This is the prior).
- (b) Does playing the game 5 times in one day satisfy the criteria for a Binomial distribution?
- (c) Suppose the game is on the hard setting. Give the conditional pmf for the number of times out of 5 you will beat the game. Repeat the process for the medium and easy settings.
- (d) Now suppose you win 4 of the 5 games you play. How likely is it that the game was set on hard? Medium? Easy? (This is the likelihood function).
- (e) Now calculate the marginal probability that you would win 4 of the 5 games. (This is the normalizing constant.)
- (f) Using the results above, you can now calculate the probability that π is a particular value, given that you won 4 of the 5 games. (This is the posterior.)

Bayes' Rule for Random Variables

Example 4. Suppose you calculate the following. Use the shortcut to derive the posterior probability $f(\pi = .5|y = 5)$:

$$f(.2)L(.2|5) = .07$$

$$f(.4)L(.4|5) = .09$$

$$f(.5)L(.5|5) = .04$$

Proportionality

Example 5. The geometric distribution models the number of independent trials Y until a success occurs. The pmf of Y conditional on a success probability π is $f(y) = \pi(1 - \pi)^{y-1}$. Flipping a weighted coin until you get a heads is an example of such a distribution. Consider the following prior pmf on the probability of a heads. Calculate the posterior pmf of π conditional on getting a heads on your 4th coin flip.

π	.1	.2	.4	.7
$f(\pi)$	.2	.3	.4	.1